

1. Create a python script file called network_automation.py (20 points)

For the purpose of this lab, we will be hardcoding passwords in the code. Do not do this in production. You should never hardcode passwords for devices you care about.

- a. Create a dictionary called rt1_cisco
 - i. Dictionary should include the correct information for netmiko to connect to the device with an ssh key (no password, make up the path to the key file)
 - 1. I don't care about what you set the value of the keys to, just looking to ensure you understand how to define devices for netmiko to connect to
 - 2. Ensure you include the enable password/secret
 - ii. Operating system to base the device type off of is Cisco IOS
- b. Create a dictionary called rt2_cisco
 - i. Dictionary should include the correct information for netmiko to connect to the device with an ssh key (no password)
 - 1. I don't care about what you set the value of the keys to, just looking to ensure you understand how to define devices for netmiko to connect to
 - 2. Ensure you include the enable password/secret
 - ii. Operating system to base the device type off of is Cisco IOS
- c. Create a dictionary called sw1_hp
 - i. Dictionary should include the correct information for netmiko to connect to the device with username and password
 - 1. I don't care about what you set the value of the keys to, just looking to ensure you understand how to define devices for netmiko to connect to
 - ii. Operating system to base the device type off of is Cisco IOS
- d. Add rt1_cisco and rt2_cisco to a list called "rt_list"
- e. Create a list called "rt_commands" with the following commands:
 - i. show version
 - ii. ip route 192.168.100.0 255.255.255.0 10.10.10.10
 - iii. show ip route
- f. Create an empty list called "sw_commands"
- g. Using a for loop, do the following:
 - i. For loop should cycle through the numbers 2-6
 - ii. For each number, generate the following command and append it to the "sw_commands" list:
 - 1. "vlan <number>"
 - a. Where <number> is the current number in the for loop (2-6)

- b. So if I was on the loop where the number was 3, “vlan 3” should be appended to the list
 - 2. “vlan <number> username<number>”
 - a. Where <number> is the current number in the for loop (2-6)
 - b. So if I was on the loop where the number was 3, “vlan 3 username3” should be appended to the list
 - 3. “tagged <number>”
 - a. Where <number> is the current number in the for loop (2-6)
 - b. So if I was on the loop where the number was 3, “tagged 3” should be appended to the list
- h. Using a for loop, loop through the routers in the list “rt_list” and do the following:
 - i. Connect to the router using netmiko
 - ii. Send the list of configuration commands in list “rt_commands”
 - iii. Print the return of sending the commands
 - iv. Disconnect from the device before moving to the next one
- i. Connect to the switch (sw1_hp) using netmiko and do the following:
 - i. Send the list of configuration commands in list “sw_commands”
 - ii. Print the return of sending the commands
 - iii. Disconnect from the device

BONUS (20 points)

Create a python script file called bonus.py

1. Create an empty dictionary called “device”
2. Create a list called “supported_devices” and populate it with the device_types related to “cisco” from the “CLASS_MAPPER_BASE” dictionary here: https://github.com/ktbyers/netmiko/blob/develop/netmiko/ssh_dispatcher.py
 - a. There should be 11 in total that are prefixed with “cisco”
3. Using a while loop:
 - a. Prompt the user to enter a device type or ‘q’ to quit
 - b. If they enter ‘q’, exit the application completely
 - i. If unsure how... Look at the “sys” package and/or the juniper.py example from class
 - c. If the device isn’t in the “supported_devices” list, tell the user the device isn’t support and print the list of supported devices
 - i. Make sure the loop continues until the device_type is supported or the user types ‘q’ to quit
 - d. If the device_type is supported, exit the loop
4. Add the provided device type to the “device” dictionary under the key “device_type”
5. Prompt the user for a username and add it to the “device” dictionary under the key “username”

6. Prompt the user for a password using the getpass library and add it to the “device” dictionary under the key “password”
7. Ask the user if they want to enter the enable password
 - a. If yes:
 - i. Prompt the user for the enable password using the getpass library and add it to the “device” dictionary under the key “secret”
 - b. If no:
 - i. Move on
8. Prompt the user for a host and add it to the “device” dictionary under the key “host”
9. Create an empty dictionary called “commands”
10. Using a while loop:
 - a. If commands is empty, prompt the user for a command and append it to the commands list
 - b. If commands isn’t empty, ask the user if they have another command they want to ran
 - i. If yes, prompt the user for a command and append it to the commands list
 - ii. If no, exit the loop
11. Create a connection to the device defined by the user which is now in the “device” dictionary using netmiko
12. If an enable password was provided, enter enable mode using netmiko
13. Using your connection, run the list of commands (these could be configuration commands so make sure you use the correct method to send a list of commands that may or may not be configuration related)
14. Print the results of running the commands
15. Disconnect from the host